

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 2.1: Introduction to Salts — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Introduction to Salts
Driving Question	DRIVING QUESTION: Why did the fertiliser granules get wet and clump together overnight, the washing soda crystals shrink and form a white crust, but the table salt stay completely the same — and what does understanding salts tell us about how to prepare and use them safely in Kenya?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is a Salt? Classification and Types	Learners observed four physical salt samples (table salt, CAN fertiliser granules, copper sulphate crystals, washing soda crystals) and recorded differences in colour, texture, crystal shape and appearance. They also observed the anchoring phenomenon — the clumped fertiliser, white-crusts washing soda and unchanged table salt — and noted that all three are salts despite behaving differently overnight.	A salt is a substance formed when a hydrogen atom in an acid is fully or partially replaced by a positive ion. Salts are classified as: normal salts (all hydrogen atoms replaced, e.g. NaCl, Na ₂ SO ₄), acidic salts (hydrogen partially replaced, e.g. NaHCO ₃ , NaHSO ₄), basic salts (contain OH ⁻ or O ²⁻ in addition to the salt anion, e.g. Pb(OH)Cl, Mg(OH)Cl), and double salts (contain two different cations or two different anions, e.g. KAl(SO ₄) ₂ ·12H ₂ O — alum). CAN fertiliser (NH ₄ NO ₃), washing soda (Na ₂ CO ₃) and table salt (NaCl) are all normal salts — yet they behave differently in air, which means classification alone does not explain the phenomenon.	Learners cannot yet fully explain WHY the three salts behaved differently overnight. They have established WHAT a salt is and HOW salts are classified, but the atmospheric behaviour (hygroscopic, deliquescent, efflorescent properties) requires understanding of solubility, water of crystallisation and vapour pressure — concepts that will be built in subsequent lessons. The DQB captures these open questions as the engine of the unit investigation.
Lesson 2 Solubility of Salts — Experiment 1	When ammonium nitrate (the salt in CAN fertiliser), sodium chloride (table salt) and sodium carbonate (washing soda) were added to water in test tubes, all three dissolved — the mixtures became clear solutions. Calcium carbonate remained as a cloudy suspension, indicating it did not dissolve. Copper sulphate dissolved and	Solubility depends on the type of salt, not its colour or appearance. All nitrates are soluble. Most chlorides are soluble. Most carbonates are insoluble except those of sodium, potassium and ammonium. Most sulphates are soluble except lead(II) sulphate and barium sulphate. These rules form the beginning of a solubility reference	We can now explain why table salt dissolves in githeri water (NaCl is soluble) and why calcium carbonate does not dissolve in river water (it is insoluble). However, we cannot yet explain why CAN fertiliser clumped overnight in the storeroom — because there was no liquid water present, only air. This means solubility in

	produced a blue solution. Lead sulphate remained cloudy and insoluble.	chart we will use throughout the unit.	liquid water is not the full explanation. Something about the air itself must be involved, and that is what we will investigate in Lesson 6.
Lesson 3 Preparing Soluble Salts I — Acid + Metal and Acid + Metal Oxide (Experiment 2a)	Zinc granules reacted with dilute sulphuric acid to produce a colourless solution and a gas that caused a burning splint to 'pop'. Black copper(II) oxide powder dissolved in warm dilute sulphuric acid to produce a blue solution. In both cases, the solid was added in excess and the unreacted solid was removed by filtration, leaving a clear salt solution.	A salt is formed when the hydrogen in an acid is replaced by a metal ion. Zinc sulphate (ZnSO_4) is prepared by the acid + metal reaction: $\text{Zn(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{H}_2\text{(g)}$. Copper sulphate (CuSO_4) is prepared by the acid + metal oxide reaction: $\text{CuO(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{CuSO}_4\text{(aq)} + \text{H}_2\text{O(l)}$. Both ZnSO_4 and CuSO_4 are soluble salts — all sulphates are soluble except BaSO_4 , PbSO_4 , and slightly soluble CaSO_4 . Excess metal or metal oxide is added to ensure all the acid is consumed, giving a pure salt product. Hydrogen gas is identified by the 'pop' test with a burning splint. The preparation method used for a salt depends on its solubility.	CAN fertiliser (calcium ammonium nitrate) is a soluble salt produced through acid reactions — the same family of reactions we carried out in the laboratory today. Because CAN is a soluble salt, its granules can interact with water molecules in humid air. Now that we understand HOW soluble salts like CAN are prepared (acid replaces its hydrogen with a metal ion to form a soluble salt), we are one step closer to explaining why those soluble salt granules absorb moisture from the air and clump together in the Rift Valley storeroom — while table salt (NaCl) and washing soda behave differently. Our next question is: how are salts prepared using carbonates and hydrogen carbonates — and will those methods add more evidence to our explanation?
Lesson 4 Preparing Soluble Salts II — Acid + Base and Acid + Carbonate/Hydrogen Carbonate	Learners observed: (1) Universal indicator changing from red/orange to green during neutralisation of HCl with NaOH , indicating $\text{pH} \approx 7$ at the endpoint; (2) Vigorous fizzing/effervescence when HCl was added to marble chips (CaCO_3); (3) Limewater turning milky/cloudy when CO_2 gas was bubbled through it — confirming the identity of the gas produced in the acid + carbonate reaction; (4) No visible solid precipitate formed in either reaction (both products were soluble salts in solution).	Learners learned: (1) Soluble salts can be prepared by reacting an acid with a base (neutralisation) — no gas is produced, and the endpoint is identified using an indicator ($\text{pH} \approx 7$); (2) Soluble salts can also be prepared by reacting an acid with a carbonate or hydrogen carbonate — CO_2 gas and water are also produced alongside the salt; (3) The balanced equation for neutralisation is: $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$; (4) The balanced equation for acid + carbonate is: $2\text{HCl(aq)} + \text{CaCO}_3\text{(s)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$; (5) The same type of reaction (acid + hydrogen carbonate $\rightarrow$ salt + CO_2 + water) explains why mandazi dough rises when baking soda is added; (6) NaCl (table salt) and CaCl_2 (calcium chloride) are both soluble salts but may behave differently when exposed to air — this will be investigated in Lesson 6.	Learners can now explain: (1) Why there is no visible change during neutralisation even though a chemical reaction is occurring — the salt (NaCl) and water produced are both colourless and soluble, so an indicator is needed to confirm the reaction is complete; (2) Why fizzing occurs when HCl reacts with CaCO_3 — CO_2 gas is produced as a product of the acid + carbonate reaction, and this same gas is responsible for the rising of mandazi dough; (3) How to write and balance chemical equations for both preparation routes, including correct state symbols; (4) How today's evidence connects to the driving question — NaCl (prepared by neutralisation) is the table salt that did not change overnight, while CaCl_2 (prepared by acid + carbonate) is hygroscopic — like the ammonium nitrate in CAN fertiliser, it absorbs water from the air, a behaviour that may explain the clumping and will be investigated in Lesson 6.

Lesson 5 Preparing Insoluble Salts — Double Decomposition and Precipitation	When barium chloride solution and sodium sulphate solution were mixed, a dense white precipitate formed immediately. The precipitate was filtered, washed with distilled water, and collected on filter paper. The filtrate (waste solution) remained clear. An alternative experiment using lead(II) nitrate and potassium iodide produced a vivid golden-yellow precipitate of lead(II) iodide.	Insoluble salts cannot be prepared by the methods used for soluble salts (acid–metal, acid–base, acid–carbonate). They are prepared by double decomposition (precipitation): two soluble salt solutions, each supplying one ion of the target salt, are mixed; the insoluble salt precipitates immediately. The net ionic equation shows only the reacting ions (e.g. $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$); the remaining ions ($\text{Na}^+$ and Cl^-) are spectator ions. The solubility of the target salt determines which preparation method must be used.	The anchoring phenomenon involves three salts behaving differently in air: CAN (ammonium nitrate — soluble, made by neutralisation), washing soda (sodium carbonate decahydrate — soluble, efflorescent), and table salt (sodium chloride — soluble, stable). Today's lesson showed that not all salts are soluble — some, like barium sulphate, are insoluble and must be made by a completely different method. This deepens our understanding of the phenomenon: the properties of a salt (solubility, stability in air, reactivity) are linked to its composition and the method by which it was formed. Choosing the right preparation method — and understanding why — is essential for producing salts safely and correctly for use in agriculture (fertilisers), medicine (barium meals at KNH), and industry across Kenya.
Lesson 6 Behaviour of Salts in Air: Hygroscopic, Deliquescent and Efflorescent Salts	Learners placed pre-weighed samples of anhydrous CaCl_2, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$, and NaCl on watch glasses and left them exposed to classroom air for 20–25 minutes. They recorded mass changes at timed intervals and observed that CaCl_2 gained mass and became moist/sticky, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ lost mass and developed a white powdery surface, and NaCl showed no significant change.	Salts behave differently when exposed to air: deliquescent salts (e.g., CaCl_2, CAN fertiliser) absorb water vapour from the air and can dissolve in it; efflorescent salts (e.g., $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$, washing soda) lose water of crystallisation to the air forming a white powder ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O} + 9\text{H}_2\text{O}$); stable salts (e.g., NaCl, table salt) show no significant interaction with atmospheric moisture under normal conditions.	The CAN fertiliser granules clumped and became wet because CAN contains deliquescent salts that absorbed water vapour from the humid storeroom air and dissolved in it. The washing soda crystals shrank and formed a white crust because $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ is efflorescent — it lost its water of crystallisation to the air. Table salt was unchanged because NaCl is stable in air. These properties determine how salts must be stored and used safely in Kenya: CAN fertiliser must be kept in sealed, airtight bags, and washing soda in sealed containers.
Lesson 7 Applications of Salts in Daily Life in Kenya	Learners read structured texts and examined images showing: CAN fertiliser being applied on Rift Valley maize farms and stored in sealed bags; sodium chloride being rubbed onto tilapia and omena at Kisumu's Dunga Beach; ORS being prepared at a rural dispensary; soda ash from Lake Magadi being used in glass manufacture; washing soda being added to laundry water; and barium sulphate being used as a pigment in paint.	The choice of a salt for a specific application depends on its type, solubility, and behaviour in air. Deliquescent salts like CAN (calcium ammonium nitrate) must be stored in sealed containers because they absorb water vapour — the same property that caused granules to clump in the phenomenon. Efflorescent salts like washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) lose water of crystallisation in dry air, forming a white powdery crust — this is why it is effective as a laundry water softener but must be stored	All three substances in the phenomenon are salts, but their different behaviours in air — and their different real-world uses — are both explained by the same underlying properties. CAN is deliquescent: it absorbs water vapour from the air, which is why granules clump overnight AND why fertiliser manufacturers and Rift Valley farmers must store it in sealed bags to prevent caking and nutrient loss. Washing soda is efflorescent: it loses water of crystallisation to the dry air, forming sodium carbonate

		in airtight containers. Insoluble salts like barium sulphate are valuable as paint pigments precisely because they do not dissolve. Soluble salts like sodium chloride and ORS components are effective in food preservation and medicine because they readily dissolve and interact with biological systems.	monohydrate ($\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$) as a white crust AND, separately, its carbonate ions make it a powerful laundry water softener by precipitating calcium and magnesium ions as insoluble carbonates. Table salt is neither hygroscopic nor efflorescent under normal conditions: it does not react with air moisture, which is why it remains unchanged in the kitchen store AND why it is a reliable, stable preservative for fish and food. The property that determines how a salt behaves in air is the same property that determines how — and whether — it can be safely and effectively used in Kenyan agriculture, food, medicine, and industry.
Lesson 8 Unit Synthesis, Model Revision and Summative Assessment	Learners observed (in Lesson 1) that CAN fertiliser granules left overnight in the school storeroom had clumped together and felt wet and sticky without any water being added; that washing soda crystals on the laboratory bench had developed a white powdery crust and appeared to have shrunk; and that ordinary table salt (NaCl) in the kitchen store was completely unchanged. All three substances are salts, yet their responses to the same air were strikingly different.	Learners have learned that: (1) Salts are formed when hydrogen atoms in an acid are fully or partially replaced by a positive ion, and are classified as normal, acidic, basic or double salts. (2) Solubility follows predictable rules — all nitrates are soluble; all sulphates are soluble except barium sulphate, lead sulphate and calcium sulphate (slightly soluble); all chlorides are soluble except lead chloride and silver chloride; all carbonates are insoluble except ammonium carbonate and Group 1 carbonates. (3) The preparation method chosen depends on the solubility of the salt: soluble salts via direct synthesis, acid-metal, acid-base or acid-carbonate/hydrogen carbonate reactions; insoluble salts via precipitation. (4) Salts behave differently in air: hygroscopic salts absorb moisture without dissolving; deliquescent salts absorb enough moisture to dissolve in it (CAN/calcium ammonium nitrate); efflorescent salts lose water of crystallisation to the air forming a white powdery crust of the monohydrate (washing soda, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O} + 9\text{H}_2\text{O}$); some salts such as NaCl are unaffected under normal conditions. (5) The choice of salt for a specific use — in Kenyan agriculture, food preservation, medicine, laundry and industry — depends	Learners can now fully explain the anchoring phenomenon: CAN fertiliser granules clumped and became wet because calcium ammonium nitrate is deliquescent (or strongly hygroscopic) — its ionic components attract water molecules from the humid storeroom air, and the absorbed water causes granules to stick together, reducing fertiliser effectiveness if not stored in sealed containers. Washing soda crystals shrank and formed a white crust because $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ is efflorescent — it spontaneously loses nine of its ten water molecules of crystallisation to the drier laboratory air, producing the monohydrate ($\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$) as a white powder on the surface. Table salt (NaCl) remained unchanged because it is neither hygroscopic, deliquescent nor efflorescent under typical Kenyan ambient humidity; its crystal lattice does not attract or release water under normal conditions. Understanding these properties — and the preparation pathways and solubility rules that underpin them — explains why farmers in the Rift Valley must seal CAN bags, why washing soda tubs must be kept closed, and why NaCl is the reliable everyday preservative used across Kenya from Mombasa fish markets to upcountry kitchens.

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ATTRIBUTION AND LICENSING

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